Practical 8: Implement Simple and Multiple Linear Regression and Evaluate the Performance

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Roll No.: 75

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = pd.read_csv('insurance.csv')
df
```

Out[2]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520
...	...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest	10600.54830
1334	18	female	31.920	0	no	northeast	2205.98080
1335	18	female	36.850	0	no	southeast	1629.83350
1336	21	female	25.800	0	no	southwest	2007.94500
1337	61	female	29.070	0	yes	northwest	29141.36030

1338 rows × 7 columns

In [3]:
```
df.head()
```

Out[3]:

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

In [4]:
```
df.shape
```

Out[4]:
```
(1338, 7)
```

In [5]:
```
df.info()
```

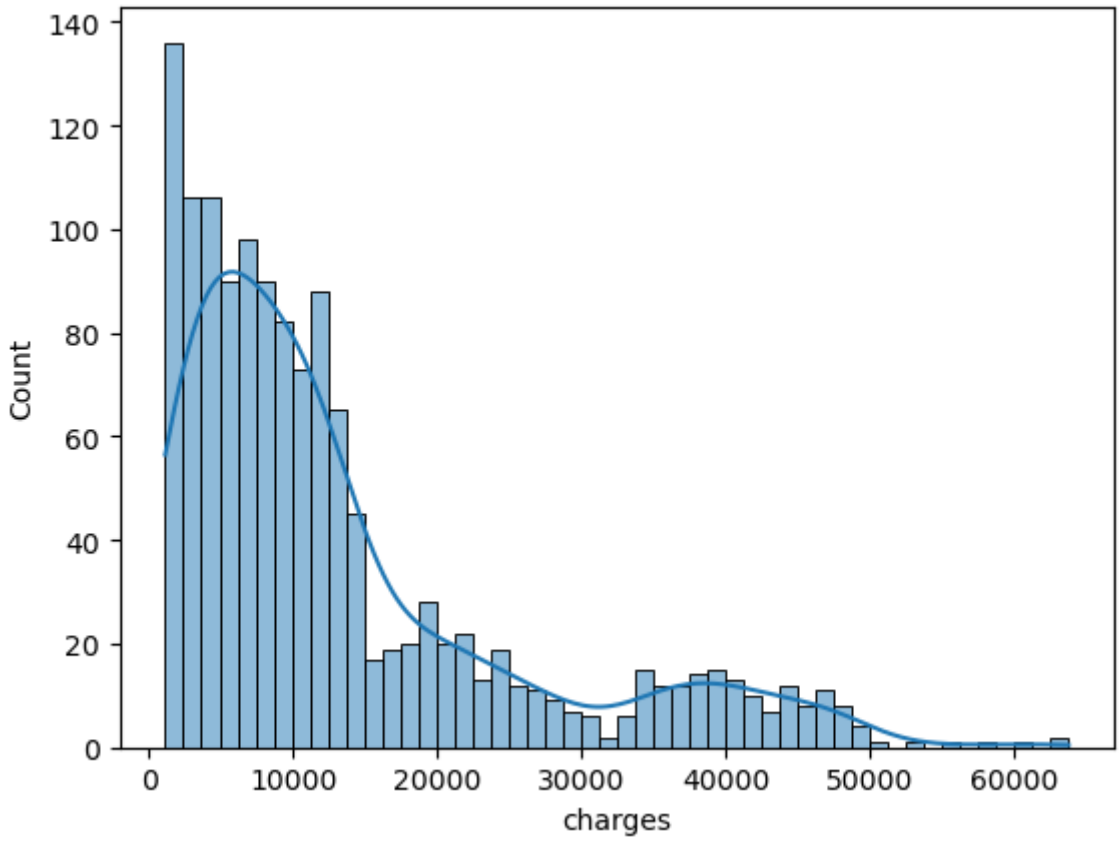
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       1338 non-null   int64  
 1   sex       1338 non-null   object 
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64  
 4   smoker    1338 non-null   object 
 5   region    1338 non-null   object 
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

In [6]:
```
df.isnull().sum()
```

Out[6]:
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

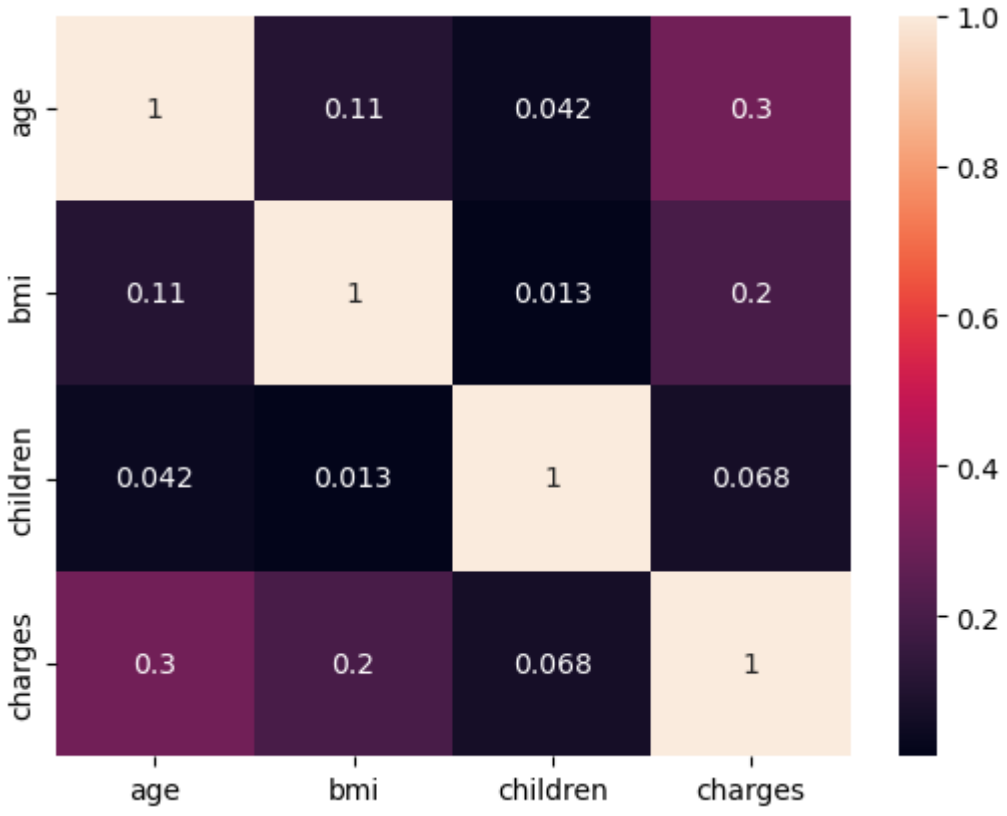
In [7]:
```
sns.histplot(df['charges'],bins = 50,kde = True)
```

Out[7]:
```
<Axes: xlabel='charges', ylabel='Count'>
```



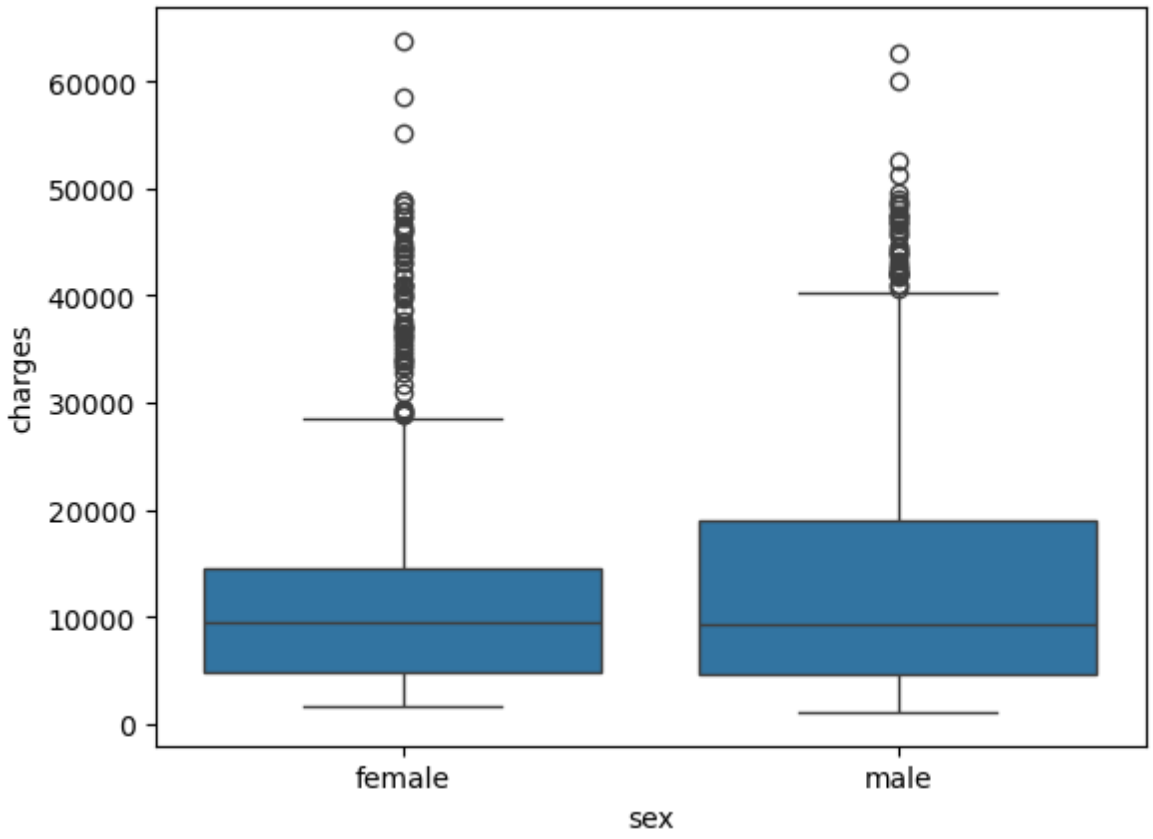
In [8]:
```
sns.heatmap(df.corr(numeric_only = True),annot = True)
```

Out[8]: <Axes: >



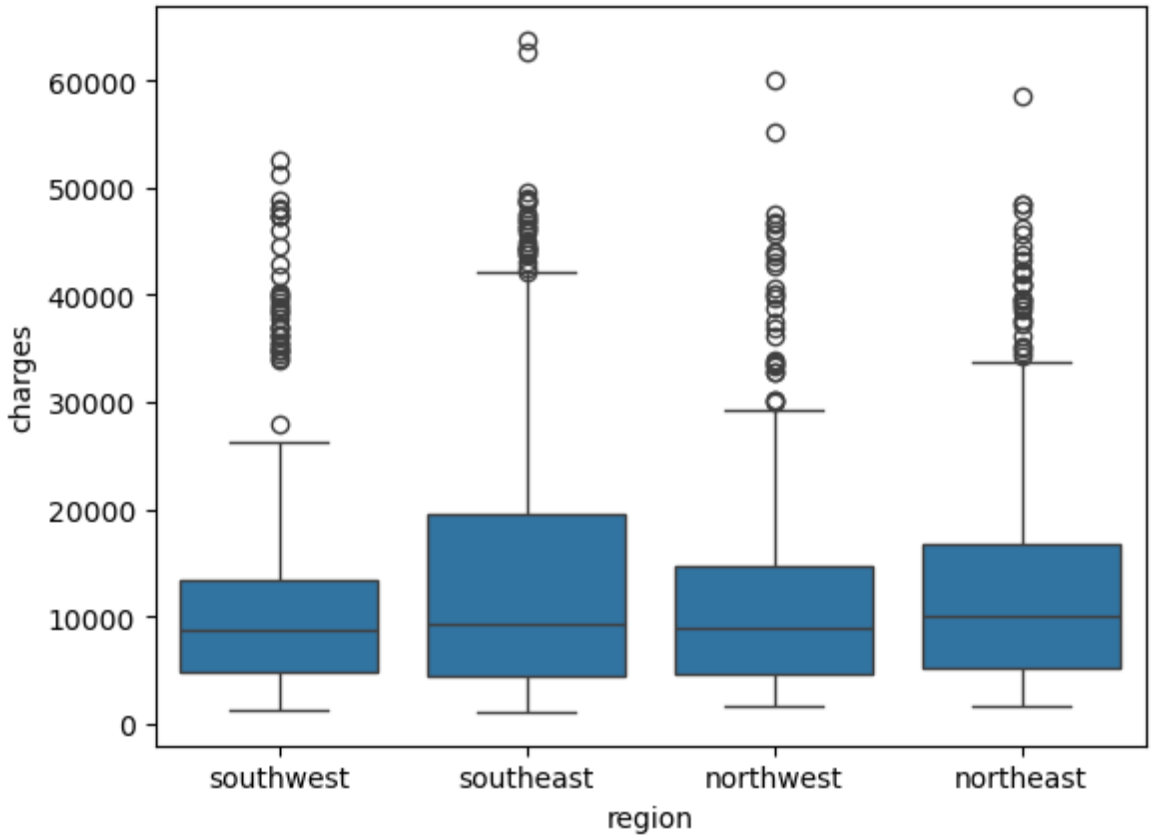
In [9]:
```
sns.boxplot(data = df,x='sex',y='charges')
```

Out[9]: <Axes: xlabel='sex', ylabel='charges'>



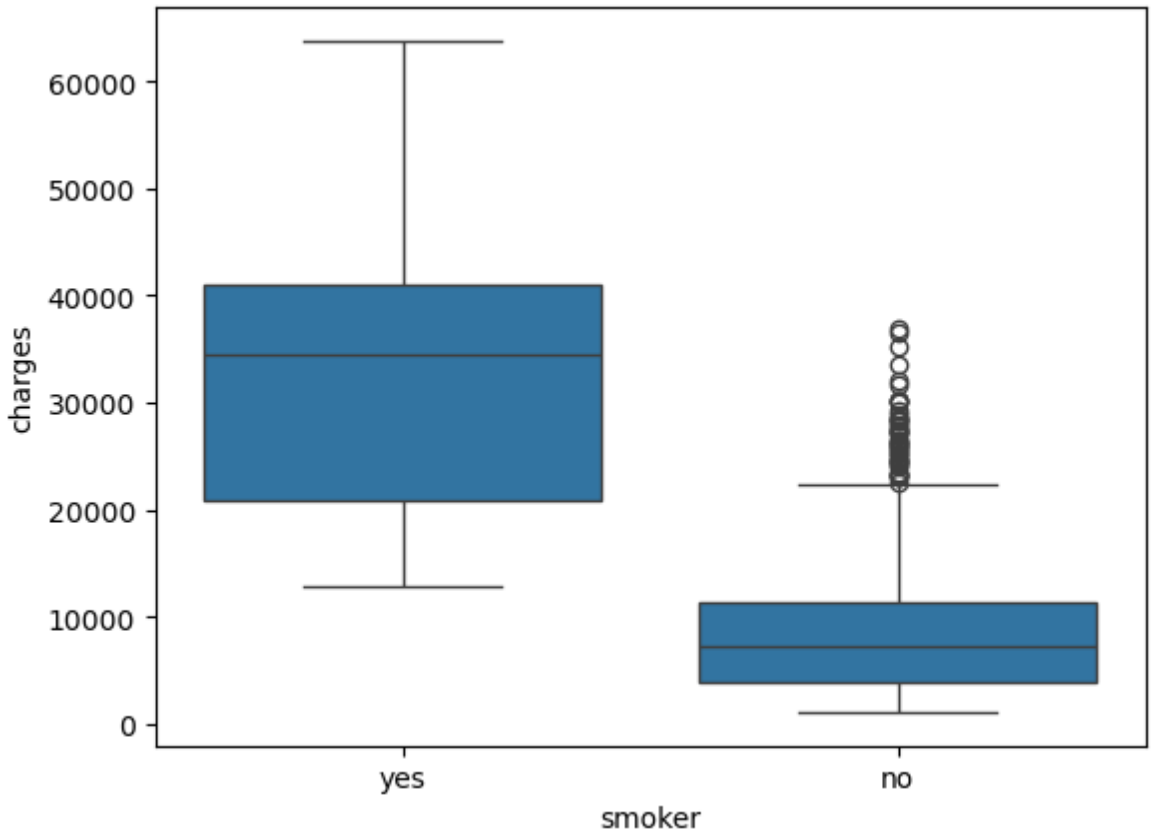
In [10]:
```
sns.boxplot(data = df,x = 'region',y = 'charges')
```

Out[10]:
```
<Axes: xlabel='region', ylabel='charges'>
```



In [11]:
```
sns.boxplot(data = df,x = 'smoker',y = 'charges')
```

Out[11]:
```
<Axes: xlabel='smoker', ylabel='charges'>
```



In [12]:
```
X = df.drop(columns = ['charges'],axis = 1)
y = df['charges']
```

In [13]:
```
X
```

Out[13]:

	age	sex	bmi	children	smoker	region
0	19	female	27.900	0	yes	southwest
1	18	male	33.770	1	no	southeast
2	28	male	33.000	3	no	southeast
3	33	male	22.705	0	no	northwest
4	32	male	28.880	0	no	northwest
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	northwest
1334	18	female	31.920	0	no	northeast
1335	18	female	36.850	0	no	southeast
1336	21	female	25.800	0	no	southwest
1337	61	female	29.070	0	yes	northwest

1338 rows × 6 columns

In [14]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
X['sex'] = le.fit_transform(X['sex'])
```

```
X['smoker'] = le.fit_transform(X['smoker'])
X['region'] = le.fit_transform(X['region'])
```

In [15]:
```
X
```

Out[15]:

	age	sex	bmi	children	smoker	region
0	19	0	27.900	0	1	3
1	18	1	33.770	1	0	2
2	28	1	33.000	3	0	2
3	33	1	22.705	0	0	1
4	32	1	28.880	0	0	1
...	...	...	...	...	...	...
1333	50	1	30.970	3	0	1
1334	18	0	31.920	0	0	0
1335	18	0	36.850	0	0	2
1336	21	0	25.800	0	0	3
1337	61	0	29.070	0	1	1

1338 rows × 6 columns

In [16]:
```
from sklearn.preprocessing import StandardScaler

numerical_cols=['age','bmi','children']
scaler = StandardScaler()
X[numerical_cols] = scaler.fit_transform(X[numerical_cols])
```

In [17]:
```
X
```

Out[17]:

	age	sex	bmi	children	smoker	region
0	-1.438764	0	-0.453320	-0.908614	1	3
1	-1.509965	1	0.509621	-0.078767	0	2
2	-0.797954	1	0.383307	1.580926	0	2
3	-0.441948	1	-1.305531	-0.908614	0	1
4	-0.513149	1	-0.292556	-0.908614	0	1
...	...	...	...	...	...	...
1333	0.768473	1	0.050297	1.580926	0	1
1334	-1.509965	0	0.206139	-0.908614	0	0
1335	-1.509965	0	1.014878	-0.908614	0	2
1336	-1.296362	0	-0.797813	-0.908614	0	3
1337	1.551686	0	-0.261388	-0.908614	1	1

1338 rows × 6 columns

In [18]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error,mean_squared_error,r2_sco
```

In [19]:
```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size = 0.33, ra
```

In [20]:
```
model = LinearRegression()
model.fit(X_train,y_train)
```

Out[20]:

▾ LinearRegression

▶ Parameters

In [21]:
```
y_pred = model.predict(X_test)
```

In [22]:
```
r2 = r2_score(y_test,y_pred)
r2
```

Out[22]:
```
0.7602640802497019
```